

Read the shape of the evidence

A decision you can explain

Connect the four branches below

1. Audit before averaging

A duplicated key may be a repeated export or a key problem. Investigate it. Missing and unusual observations need separate treatment.

2. Compare center

The mean uses every magnitude. The median uses order and position. A long right tail can pull the mean above the median.

3. Describe spread

Standard deviation summarizes variation around the mean. IQR is $Q3 - Q1$. An outlier fence is a review flag, not proof of error.

4. Choose the display

A dot plot shows each small-sample observation. A histogram depends on bins. Box plots compactly compare distributions but do not show every shape feature.

Audit before averaging

Raw rows > distinct cases > valid measurements

The connection

A duplicated key may be a repeated export or a key problem. Investigate it. Missing and unusual observations need separate treatment.

A business example

Keep a raw copy and reconciliation record. Report eligible cases and valid measurements.

Explain it without looking

Name the observational unit, the evidence needed, and one limitation. Connect this branch to the next decision in the session.

Compare center

Mean < > median < > shape

The connection

The mean uses every magnitude. The median uses order and position. A long right tail can pull the mean above the median.

A business example

Five delivery times can retain the same median while one long delivery substantially increases the mean.

Explain it without looking

Name the observational unit, the evidence needed, and one limitation. Connect this branch to the next decision in the session.

Describe spread

SD < > range < > quartiles < > IQR

The connection

Standard deviation summarizes variation around the mean. IQR is Q3 minus Q1. An outlier fence is a review flag, not proof of error.

A business example

Check a flagged value against its source. Preserve valid unusual observations and report sensitivity if it matters.

Explain it without looking

Name the observational unit, the evidence needed, and one limitation. Connect this branch to the next decision in the session.

Choose the display

Distribution > comparison > relationship

The connection

A dot plot shows each small-sample observation. A histogram depends on bins. Box plots compactly compare distributions but do not show every shape feature.

A business example

Show group sizes and missingness. A descriptive difference alone does not establish causation.

Explain it without looking

Name the observational unit, the evidence needed, and one limitation. Connect this branch to the next decision in the session.